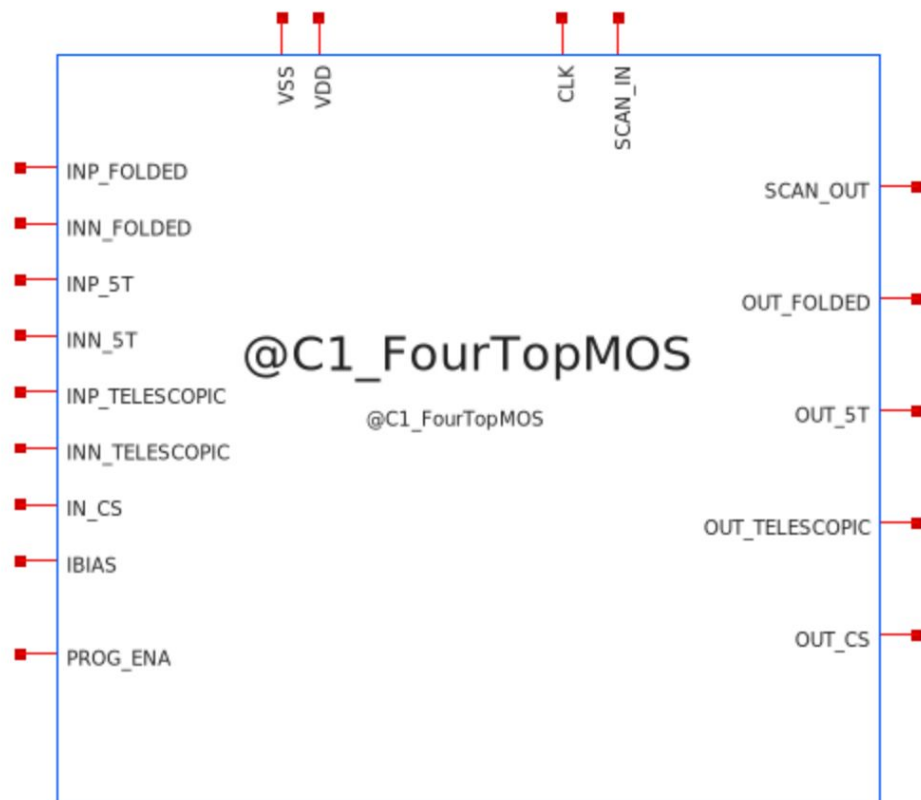


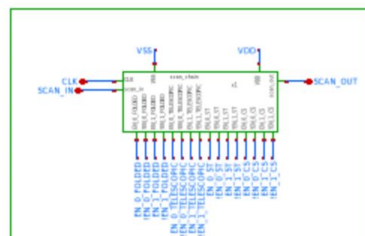
C1 Four Topology MOSbius

Four Topology MOSbius

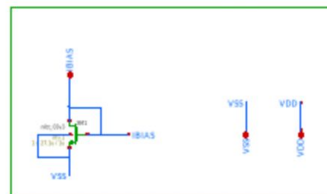
Discord name	Affiliation	Experience	Role
elij0100	Columbia University	Undergraduate	Team Lead
boibun.	Columbia University	Undergraduate	Team Member
maxdrucker_83950	Columbia University	Undergraduate	Team Member
Manwell	Columbia University	Undergraduate	Team Member



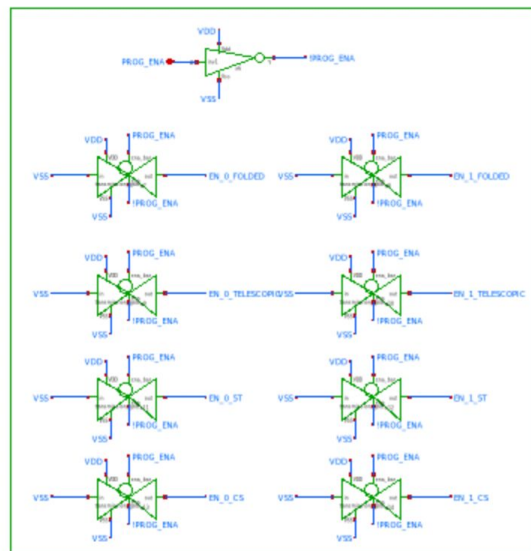
SCAN CHAIN



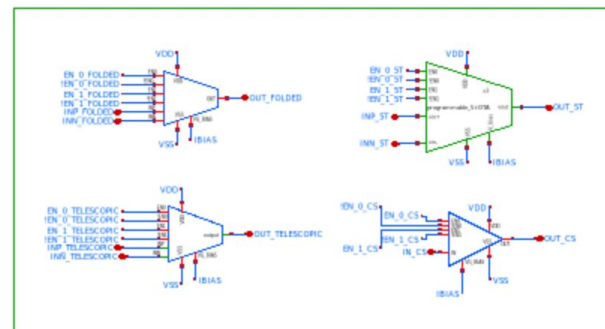
BIAS AND POWER PINS



PROGRAMMING ENABLE



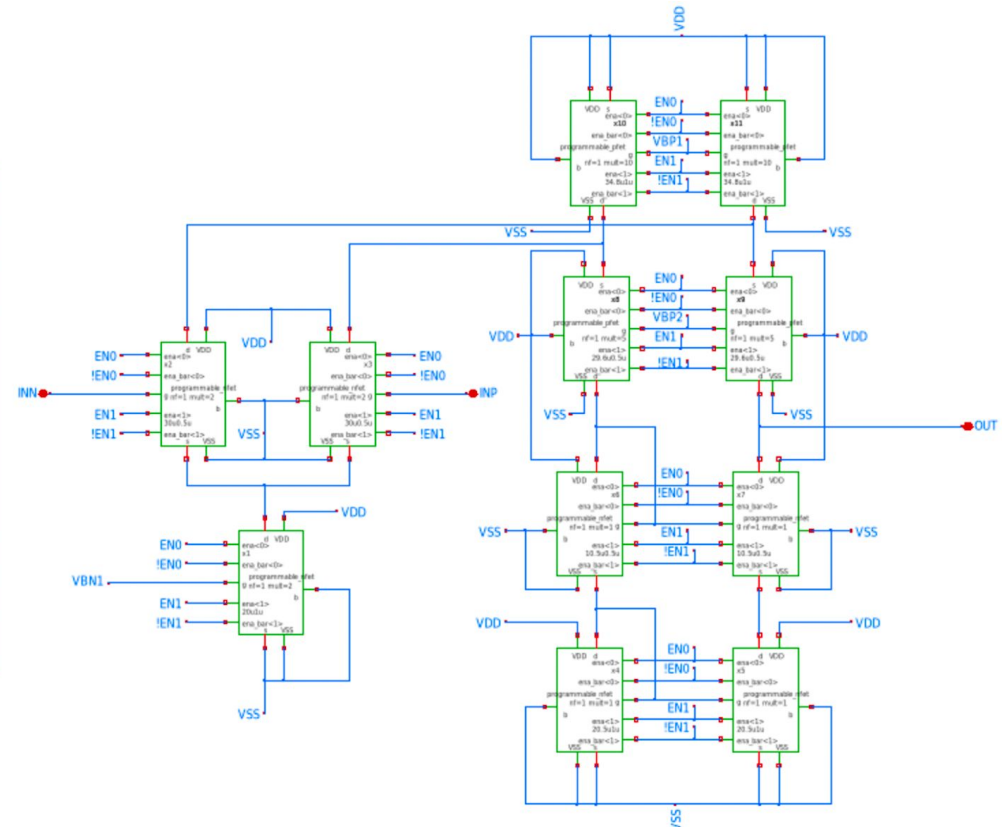
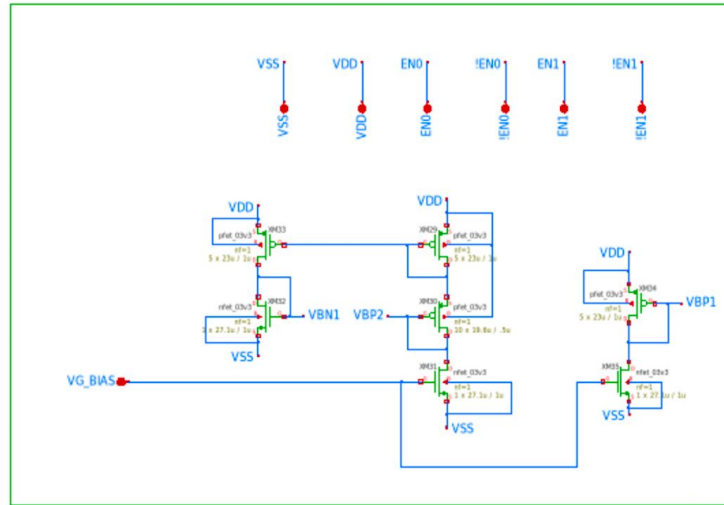
CORE AMPLIFIERS



Pull Downs on
Enable Pins



Bias Voltage Generation and Pins



Folded Cascode Sizing

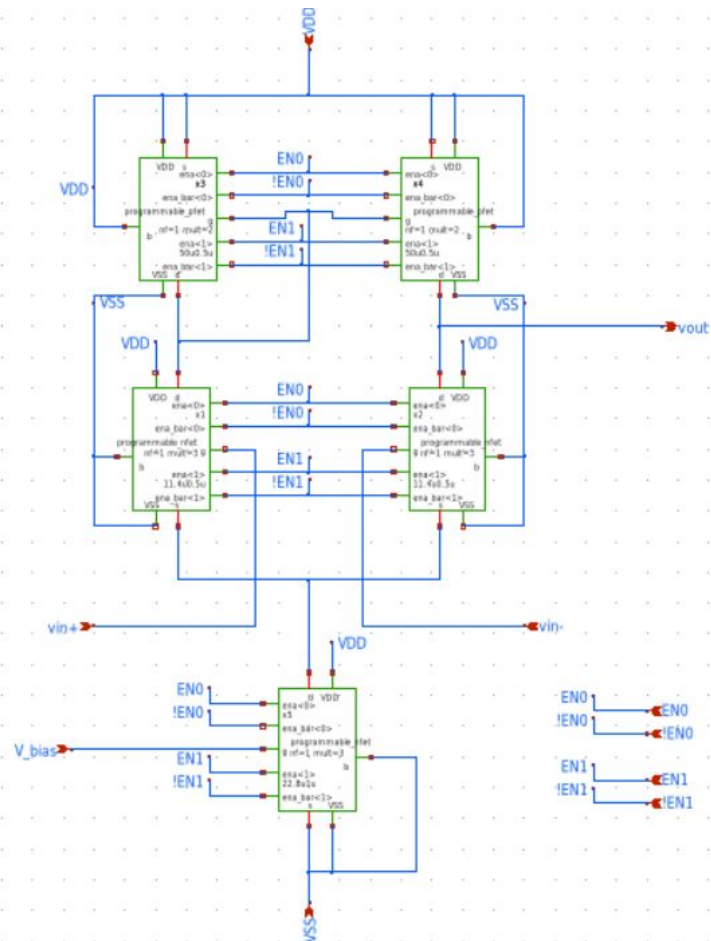
Device	Role	W (μm)	L (μm)	ID (μA)
XM10/XM11	NMOS input pair	60	0.5	75
XM12	NMOS tail	40	1	150
XM2/XM5	PMOS current sources	174	1	150
XM3/XM4	PMOS cascodes	148	0.5	75
XM6/XM7	NMOS cascodes	10.5	0.5	75
XM8/XM9	NMOS mirrors	20.5	1	75

- (Default sizing shown, for 2x, 3x, and 4x sizing, the width and bias current would scale respectively for all amplifiers)



Telescopic Cascode Sizing

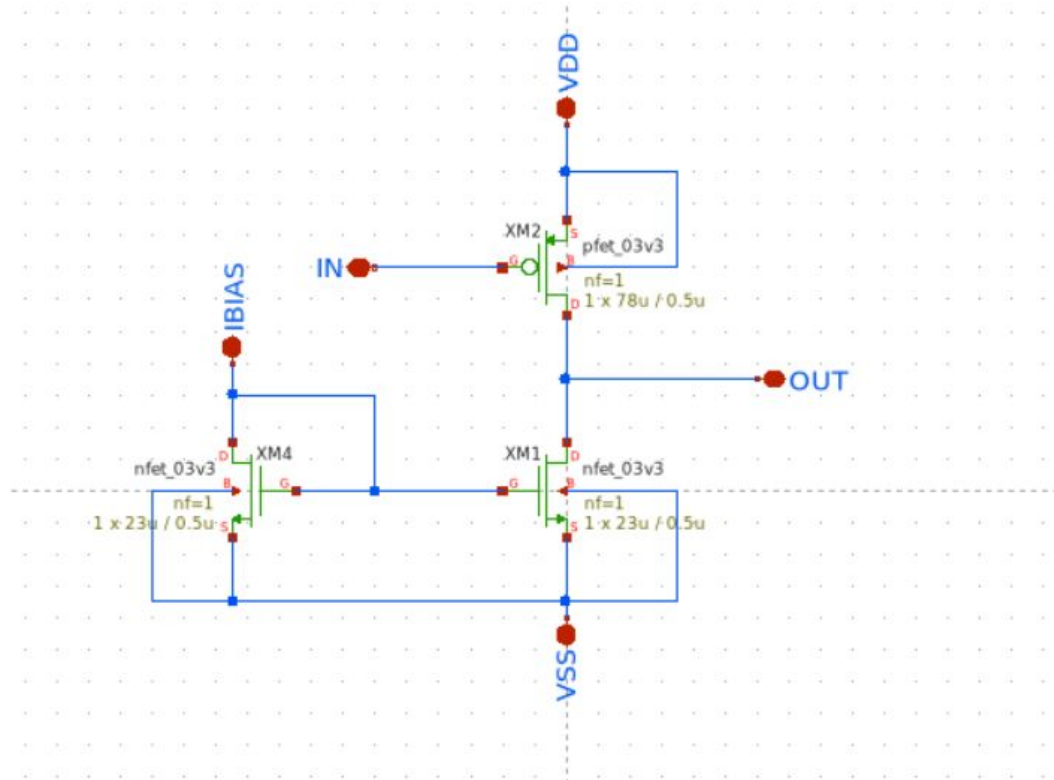
Device	Role	W (μm)	L (μm)	ID (μA)
XM1	NMOS Tail	60	0.5	200
XM2/XM3	Input Pair	30	0.5	100
XM4/XM5	NMOS Cascode	180	0.5	100
XM6/XM7	PMOS Sascodes	235	0.4	100
XM8/XM9	PMOS Current Sources	240	0.5	100



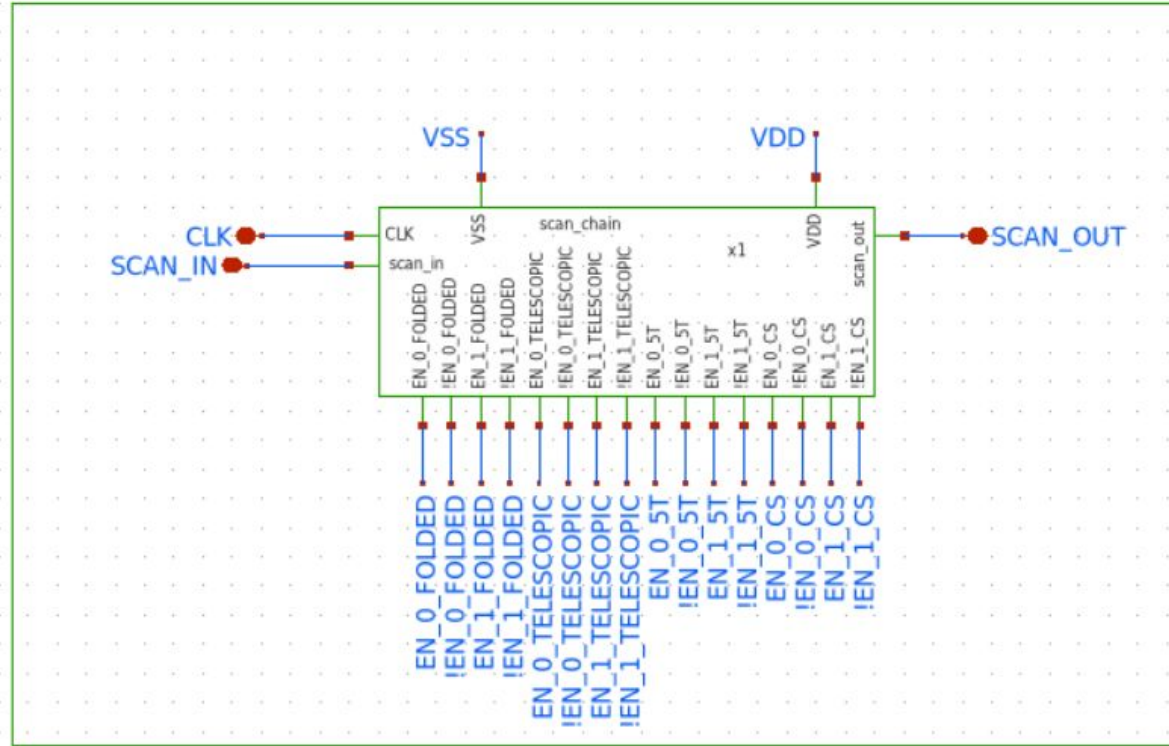
5-Transistor Sizing

Device	Role	W (μm)	L (μm)	ID (μA)
XM1/XM2	Input Pair	34.2	0.5	150
XM3/XM4	PMOS Active Load	100	0.5	150
XM5	NMOS Tail	68.4	0.5	300

Common-Source Amplifier



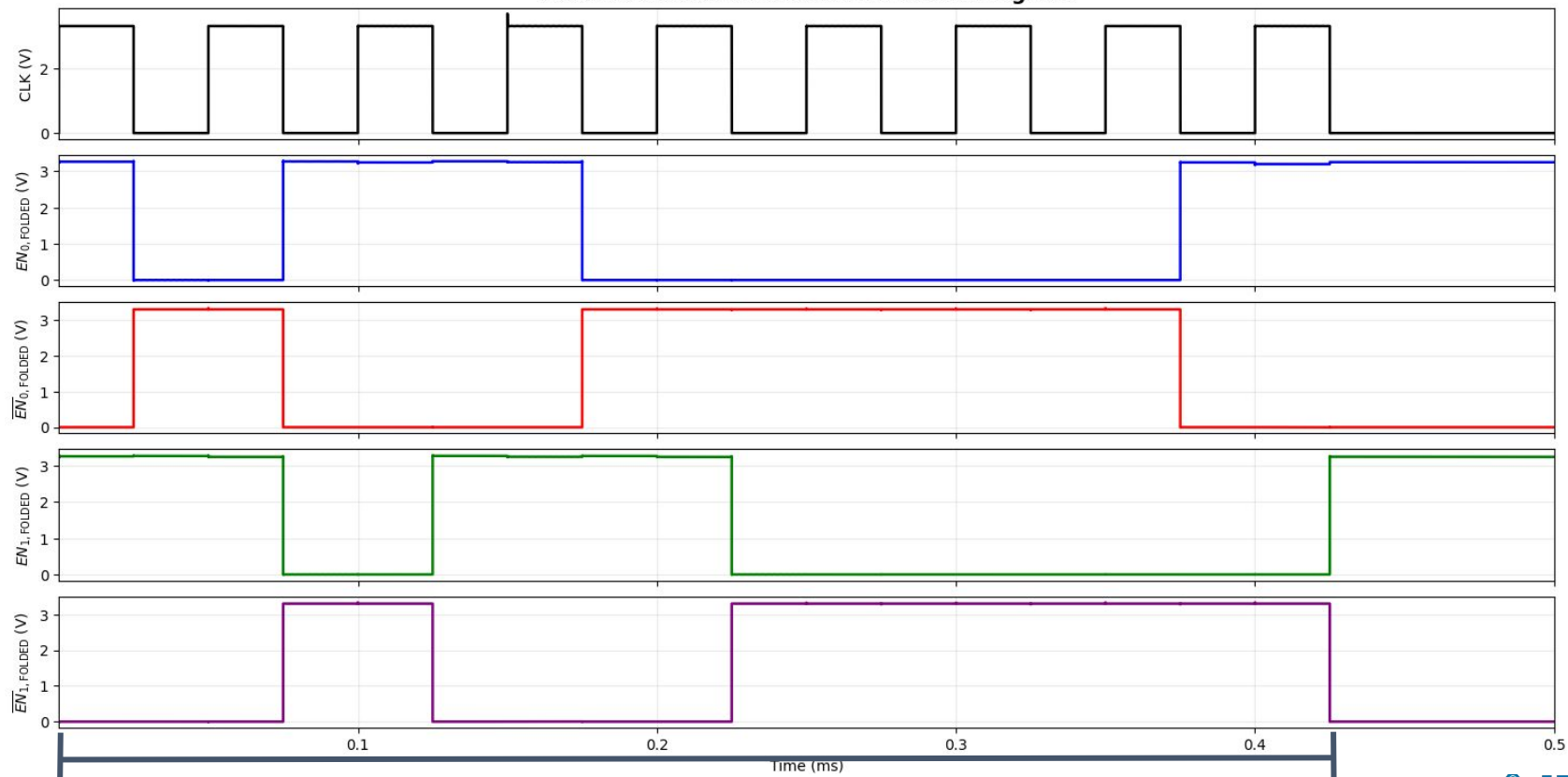
SCAN CHAIN



Clock and Scan-In Signal are to be generated by a Raspberry Pi running micropython (can also be replaced by any other microcontroller). Data for any possible configuration can be propagated in .425 ms, and verified via the Scan-Out Pin. In the next slide, the D Flip-Flop outputs of EN0 and EN1 (and compliments) are displayed, propagated in .425 ms. At .425 ms the clock turns off, and with the enable bits set in place until the user chooses to reconfigure the Four Topology MOSbius.

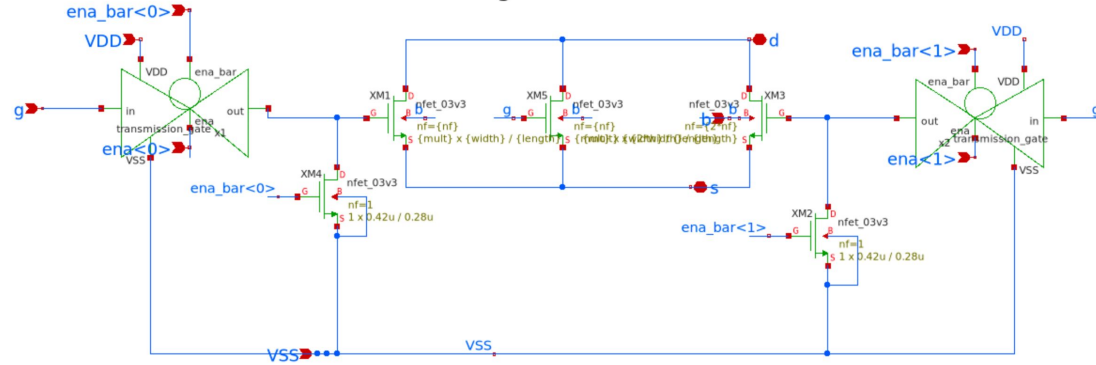
*During this data propagation, in transient simulations, you will notice voltage spikes do to digital switching, which disappear after the clock is turned off.

Folded Cascode OTA Transient Control Signals

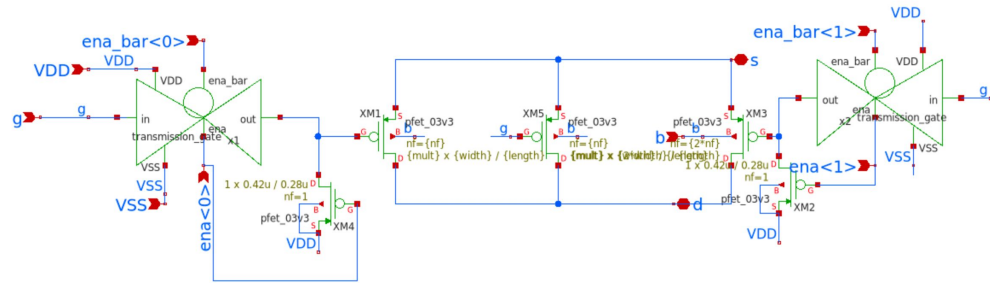


.425 ms Data Propagation

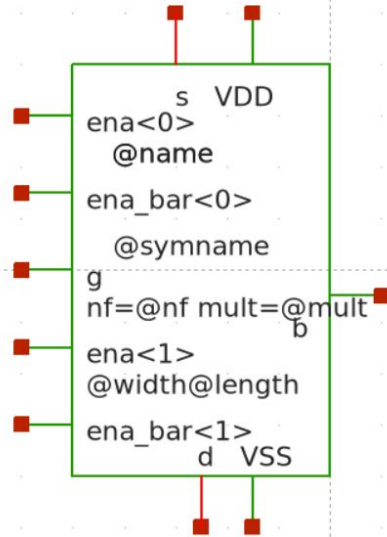
Programmable NFET



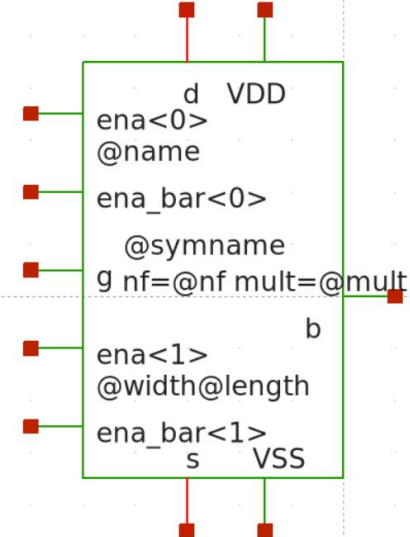
Programmable PFET



Programmable PFET



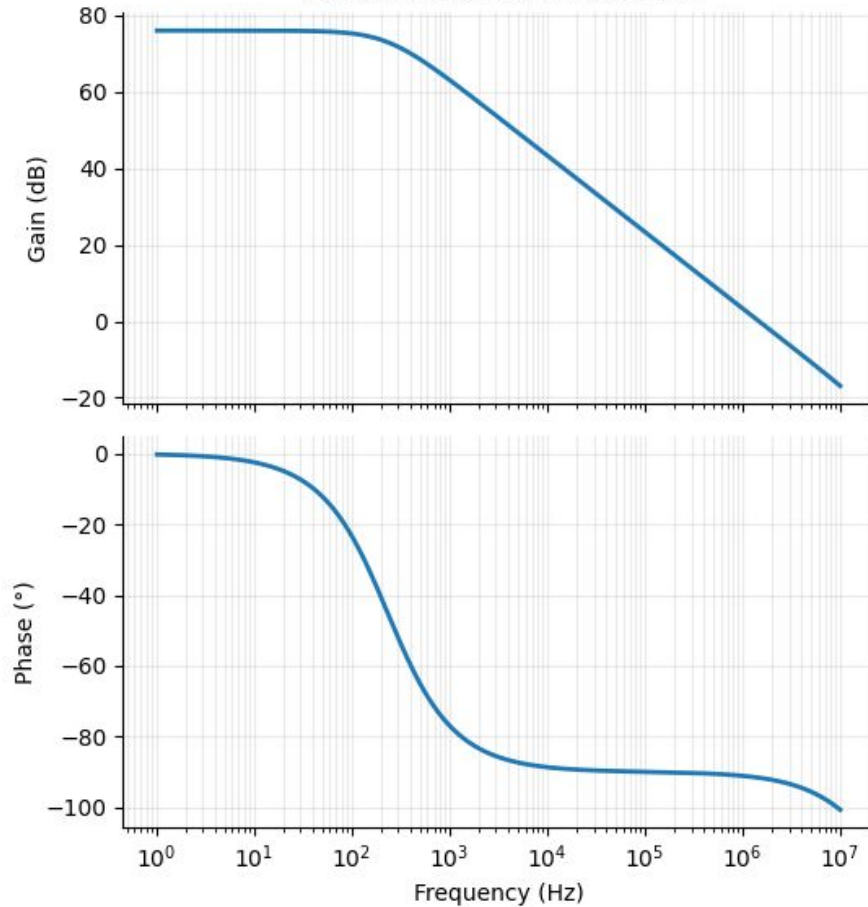
Programmable NFET



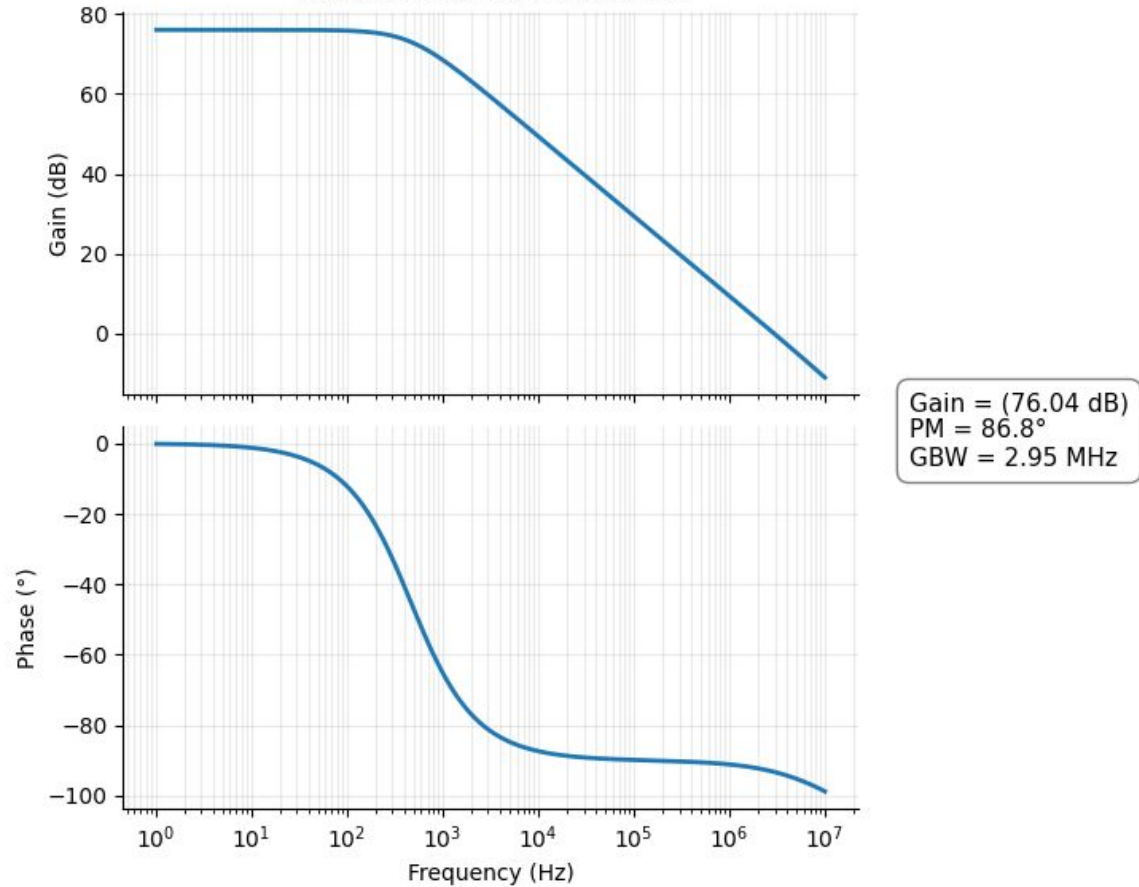
AC Simulations

The following AC Sims were extracted by hard coding the Programming Transistors (Setting the Enable Pins high or low without the need of the Scan Chain). AC sims were unable to be done directly via the Top-Level as SPICE was unable to find an operation point due to the presence of the digital platform. Instead, GBW was extracted via the GBW product. There is an error margin as result were limited by 1uS resolution simulations. These results can be found in the last slide.

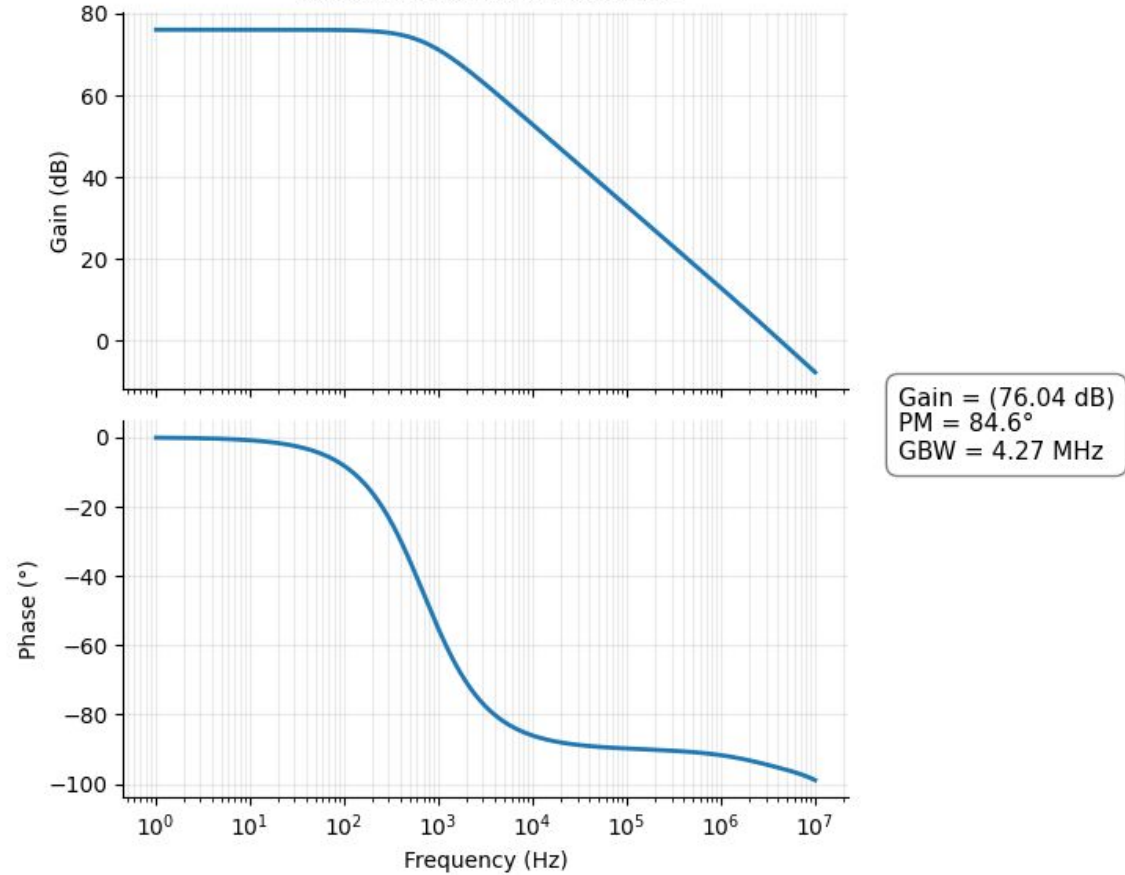
1X Folded Cascode Bode Plot



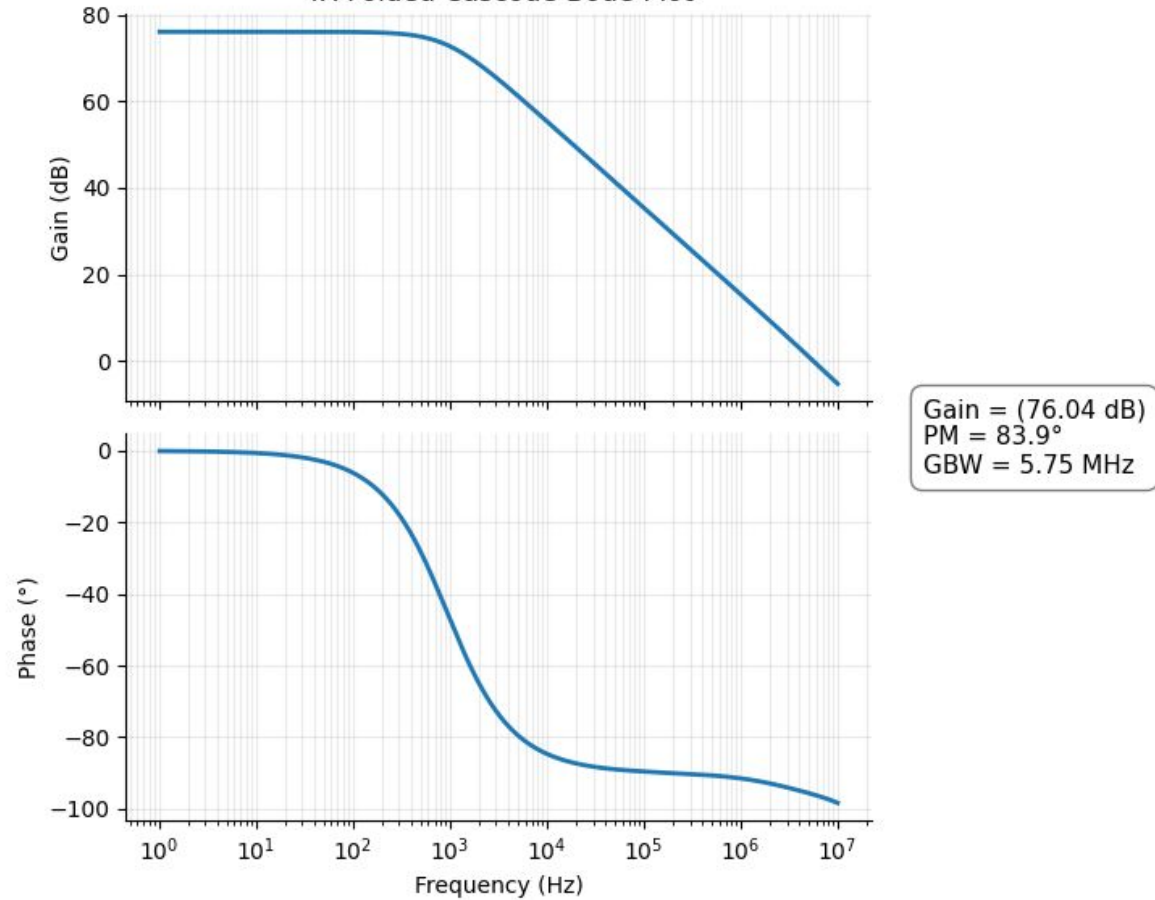
2X Folded Cascode Bode Plot



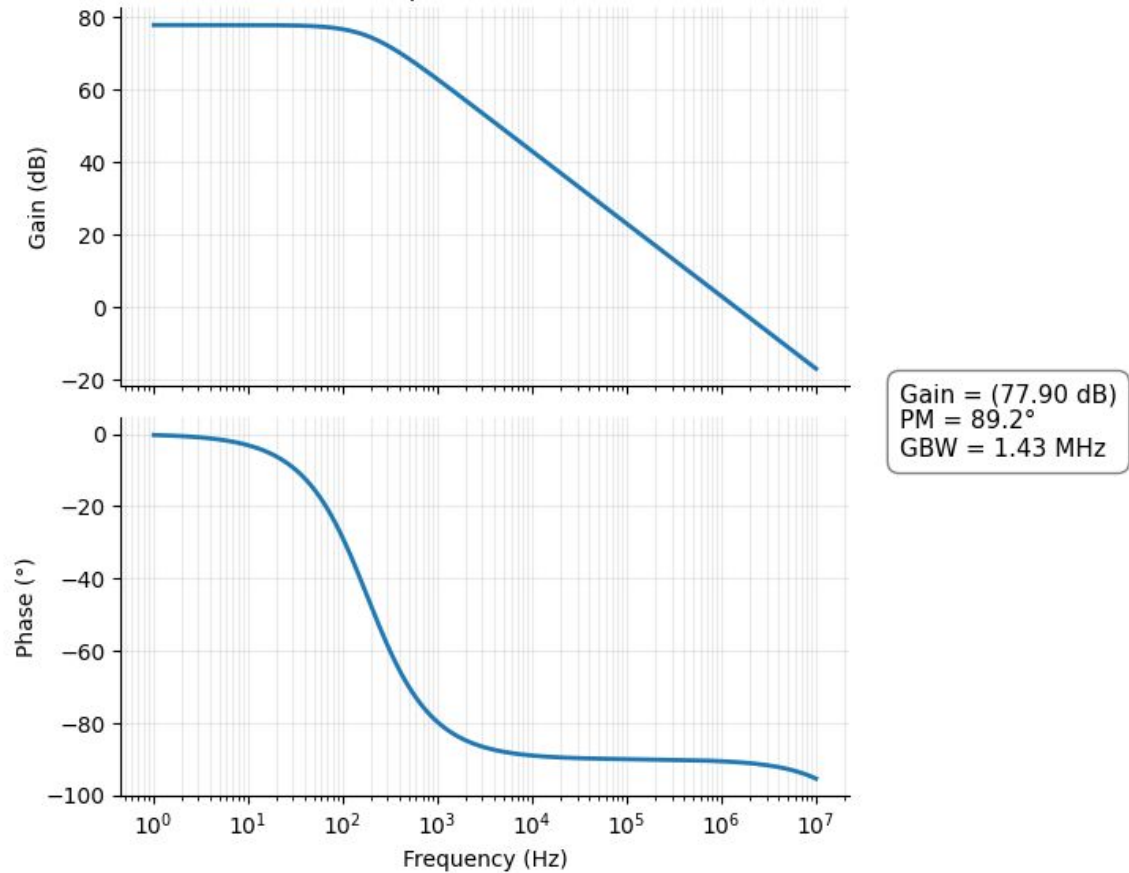
3X Folded Cascode Bode Plot



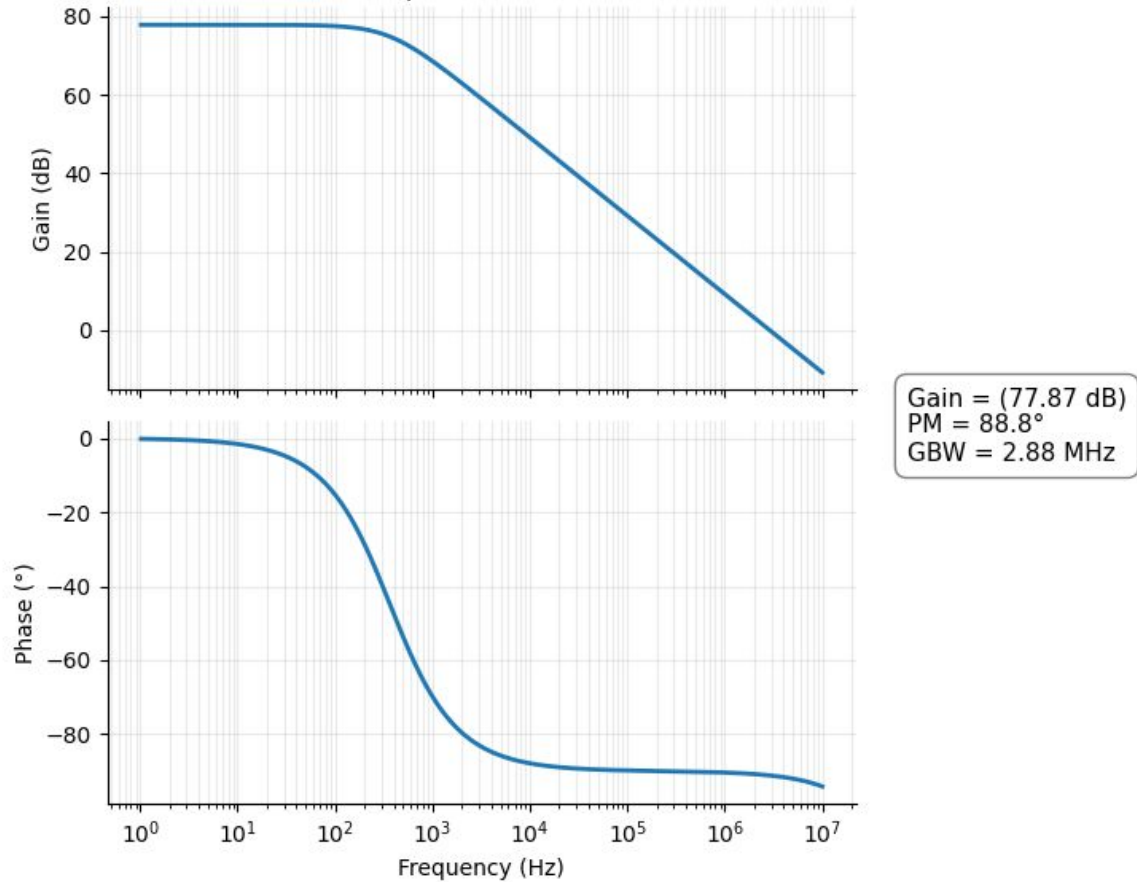
4X Folded Cascode Bode Plot



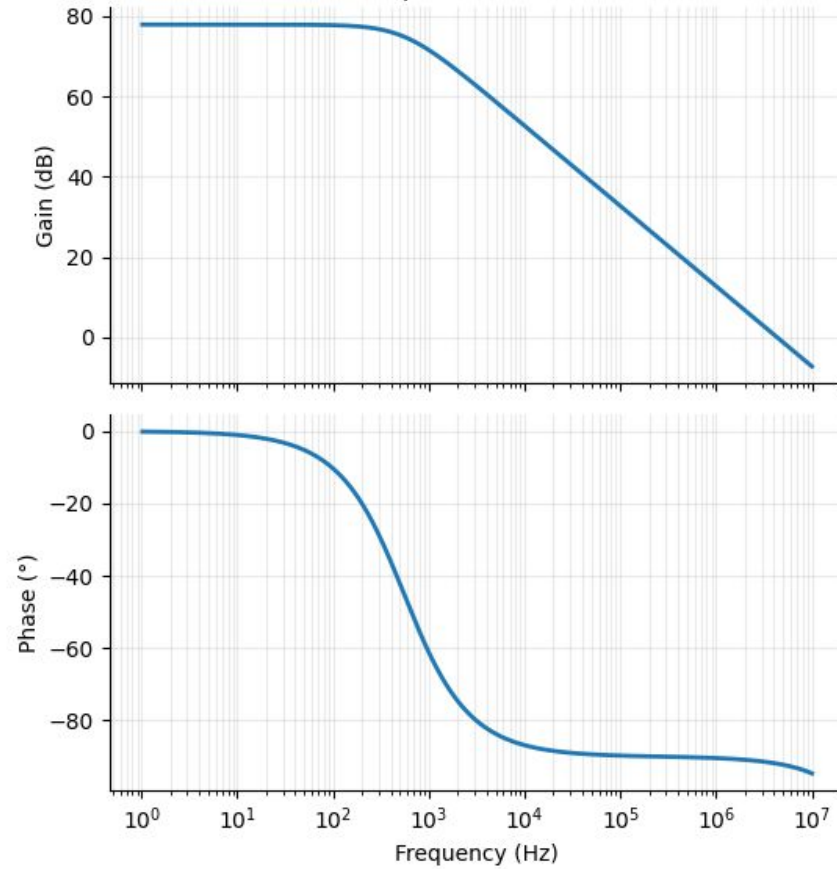
1X Telescopic Cascode Bode Plot



2X Telescopic Cascode Bode Plot

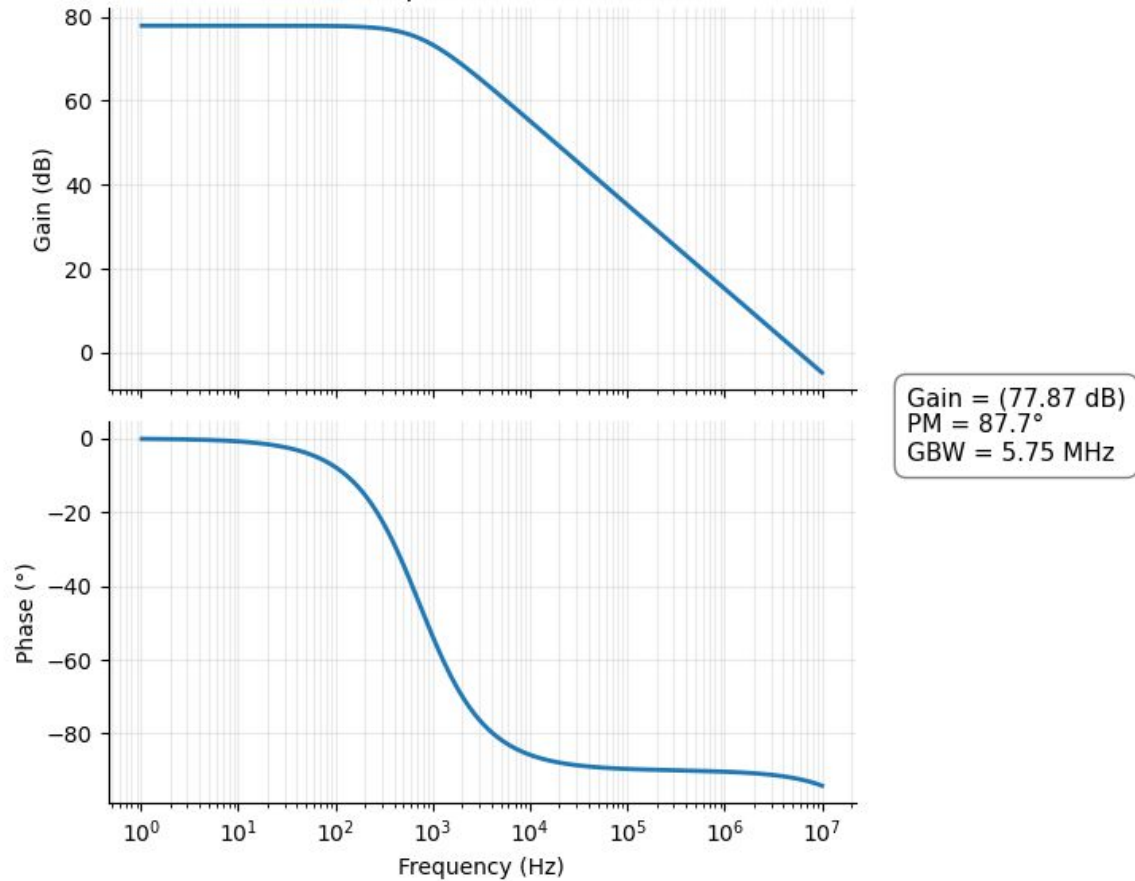


3X Telescopic Cascode Bode Plot

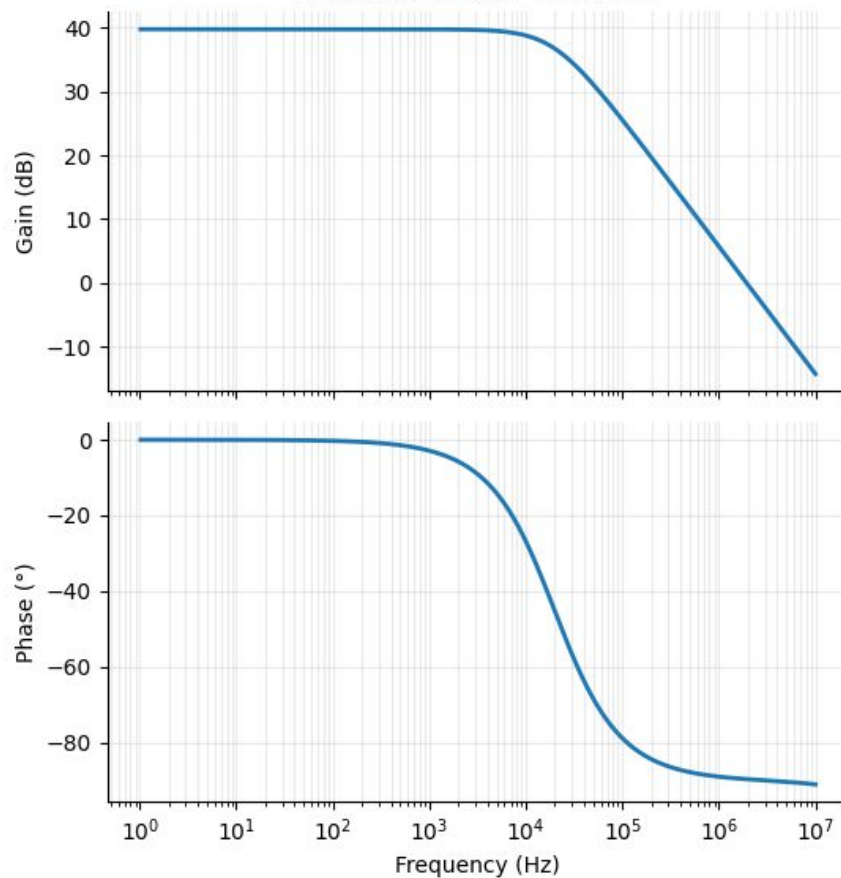


Gain = (77.87 dB)
PM = 88.0°
GBW = 4.37 MHz

4X Telescopic Cascode Bode Plot

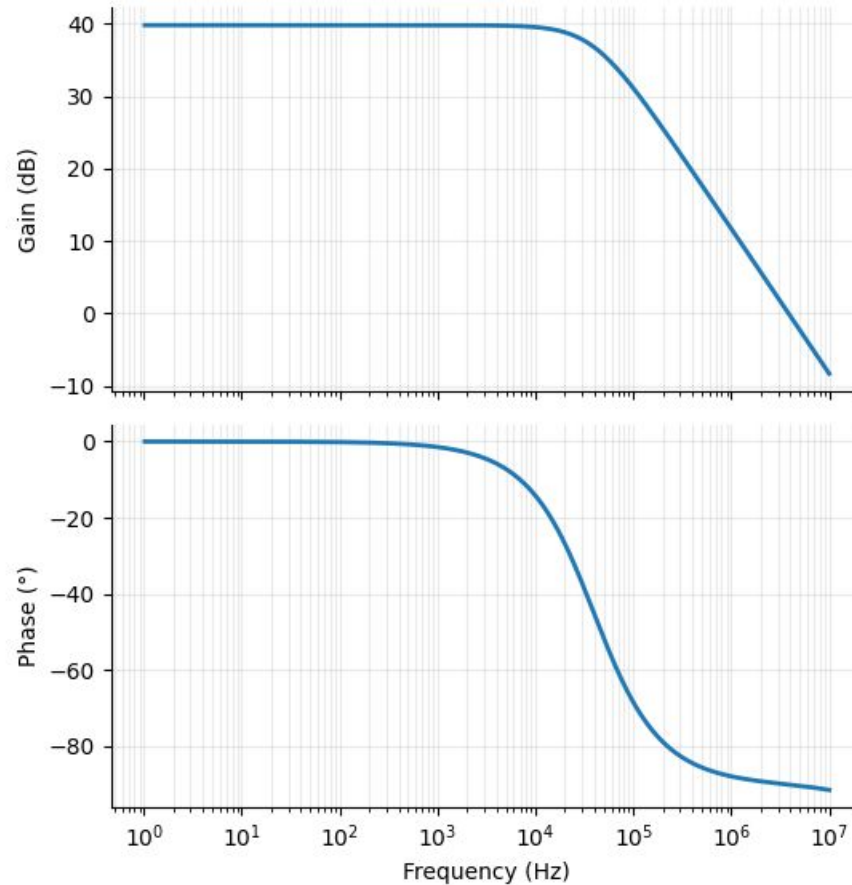


1X Five-Transistor Bode Plot



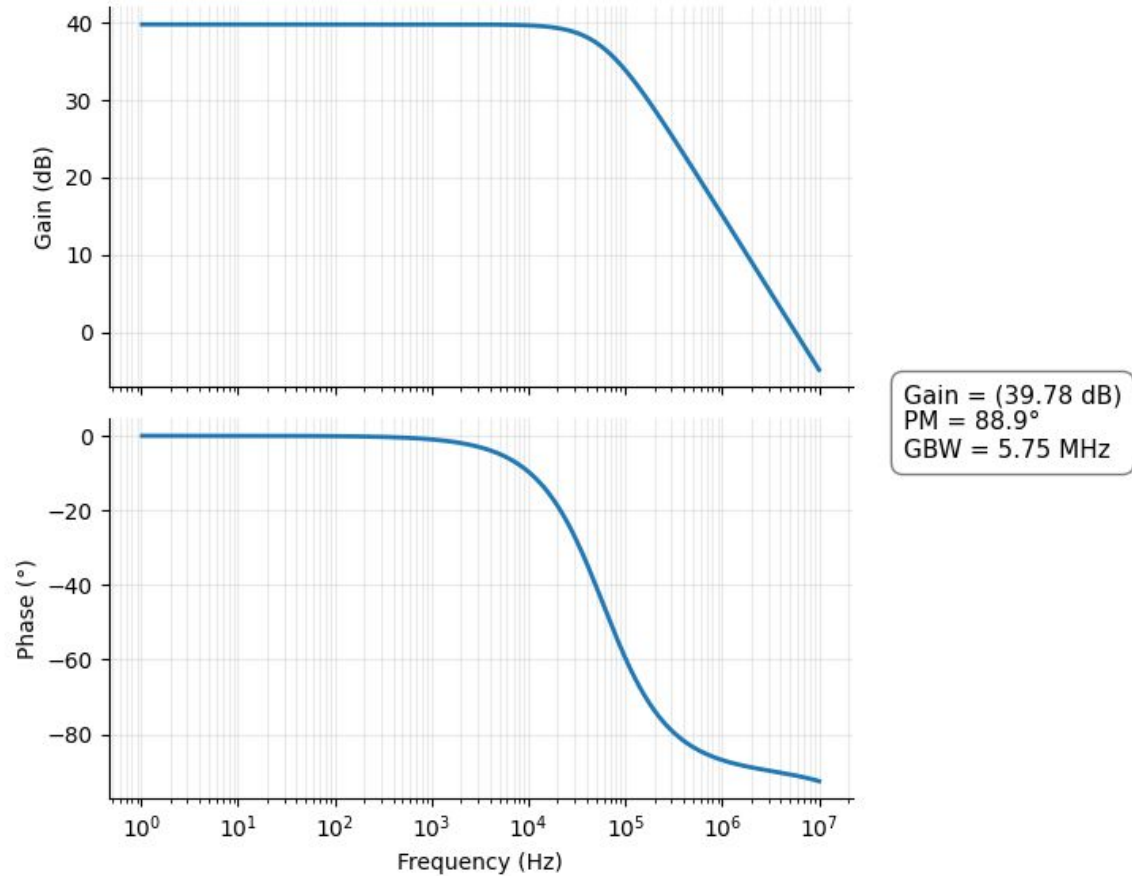
Gain = 39.78 dB
PM = 90.4°
GBW = 1.95 MHz

2X Five-Transistor Bode Plot

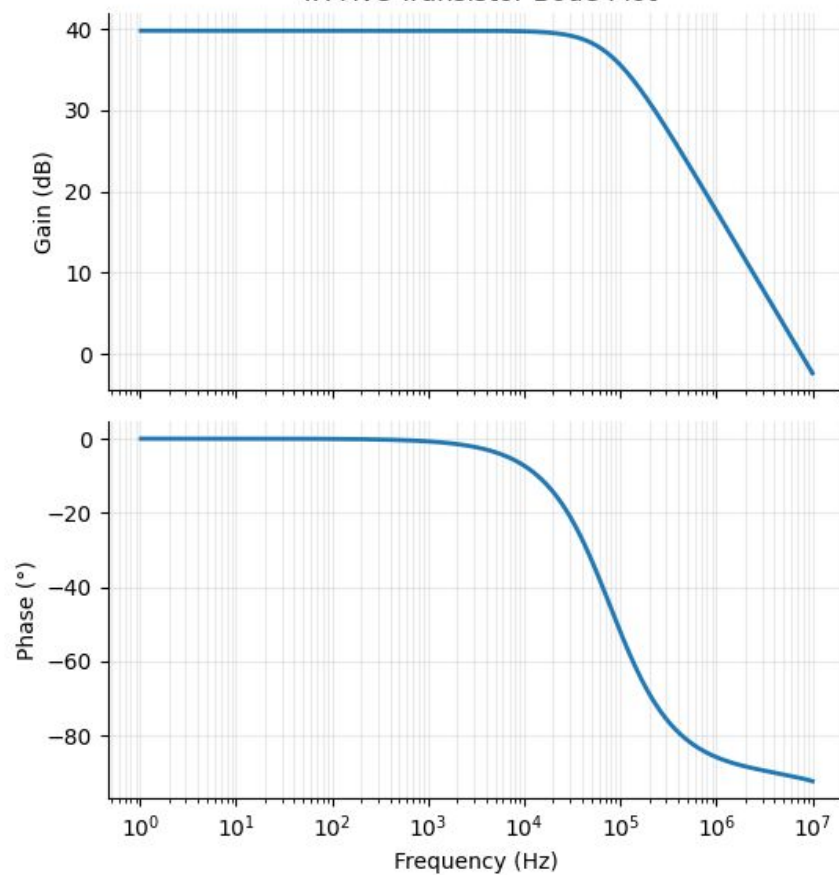


Gain = (39.78 dB)
PM = 89.9°
GBW = 3.89 MHz

3X Five-Transistor Bode Plot



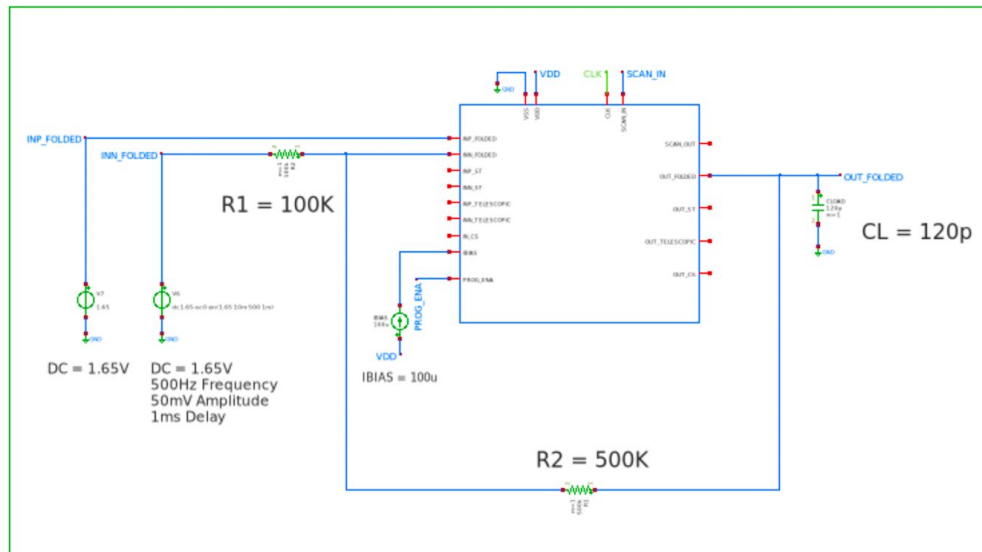
4X Five-Transistor Bode Plot



Gain = 39.78 dB
PM = 88.5°
GBW = 7.76 MHz

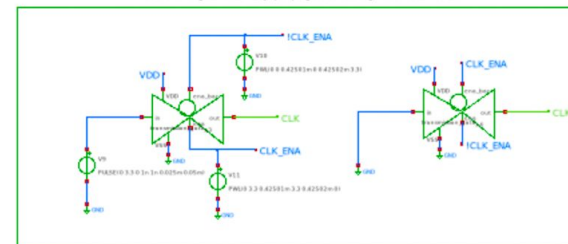
Four Topology MOSbius as an Inverting Amplifier

Configure the folded cascode in 4x mode as an inverting amplifier to amplify a sine wave input by 5 while driving a 120pF capacitive load.



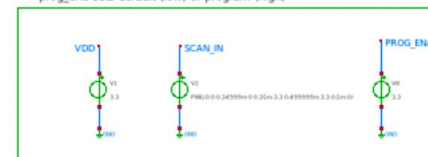
CLOCK GENERATION

Artificially generate a clock signal and ensure it turns off after around .425ms, which is when the scan in signal has fully propagated through the scan chain

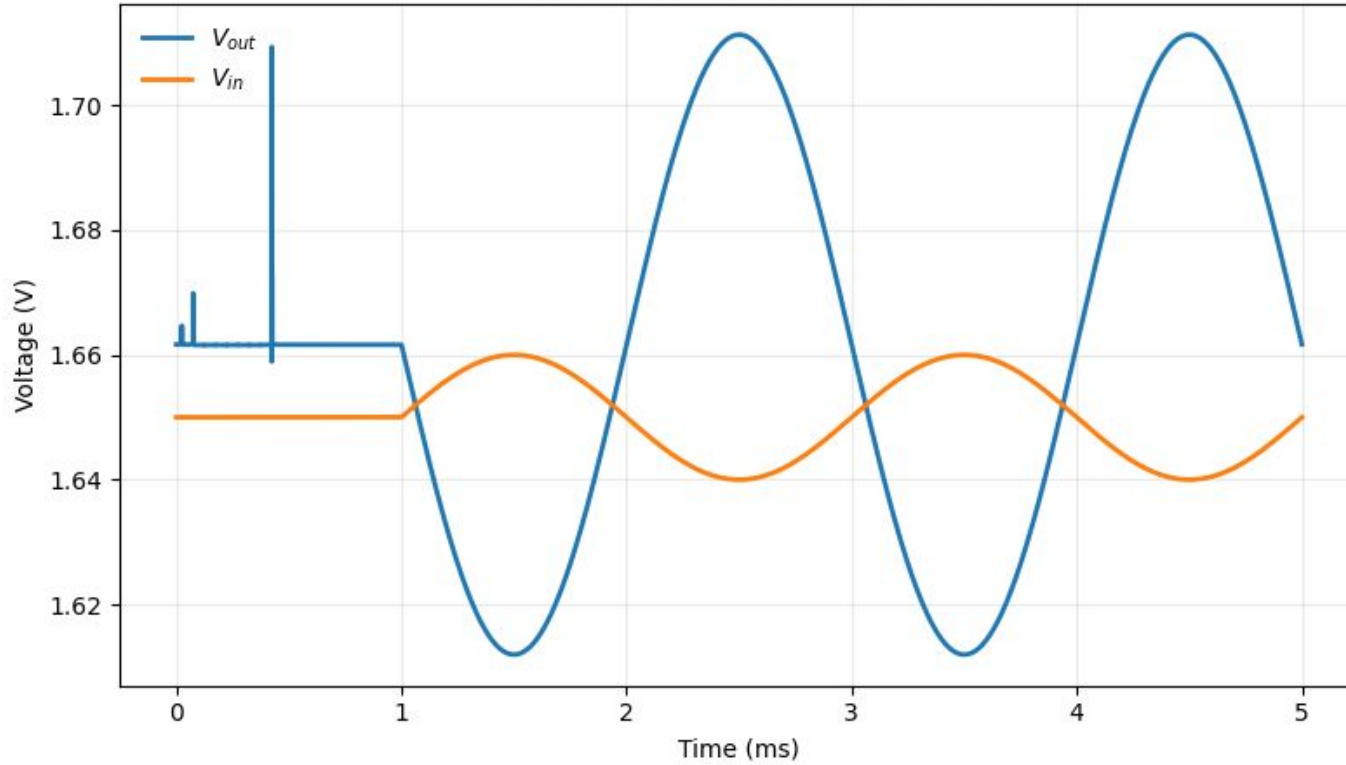


VDD, GND, SCAN INPUT, PROGRAMMABLE ENABLE

Scan chain input artificially generated to enable folded cascode 4x sizing
prog_ena sets default (low) or program (high)



Folded OTA 2X Transient Response



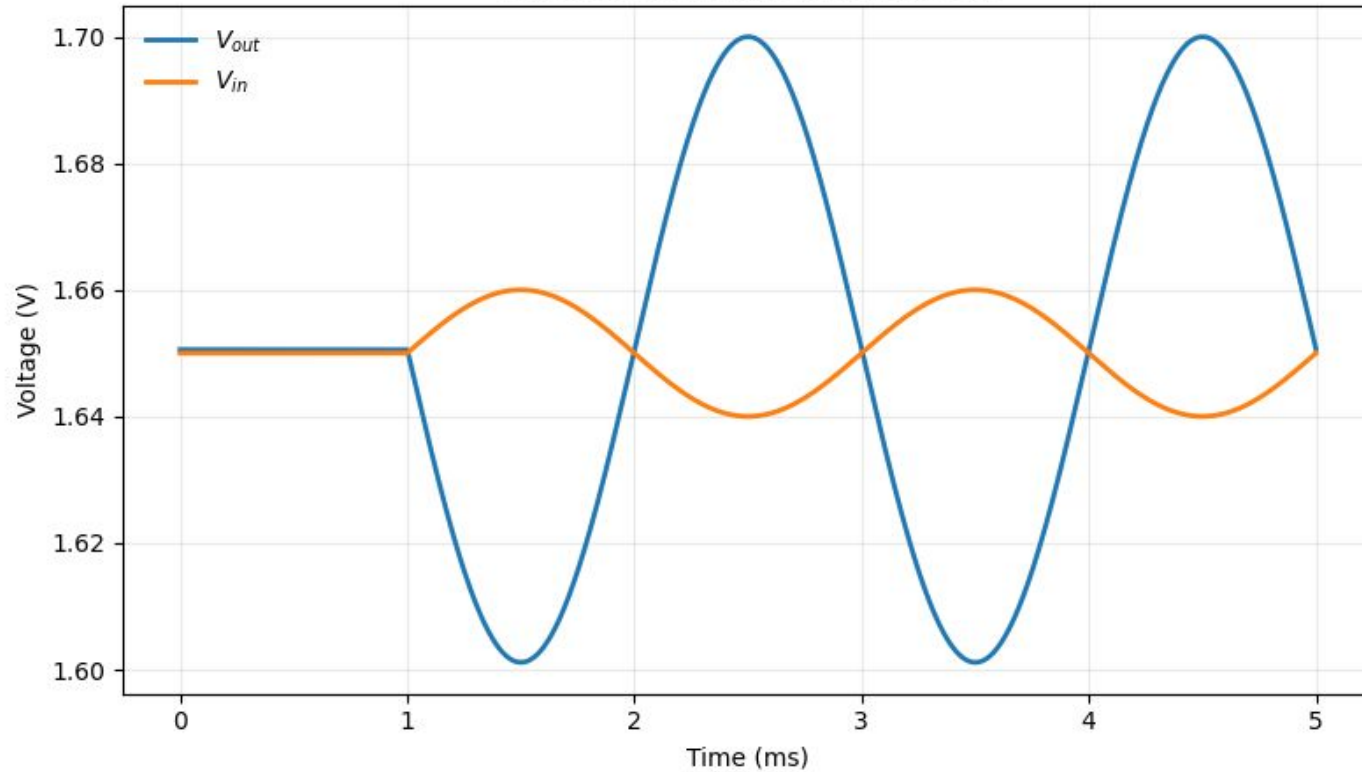
Gain

4.968728

V/V

13.92 dB

Telescopic OTA Transient Response



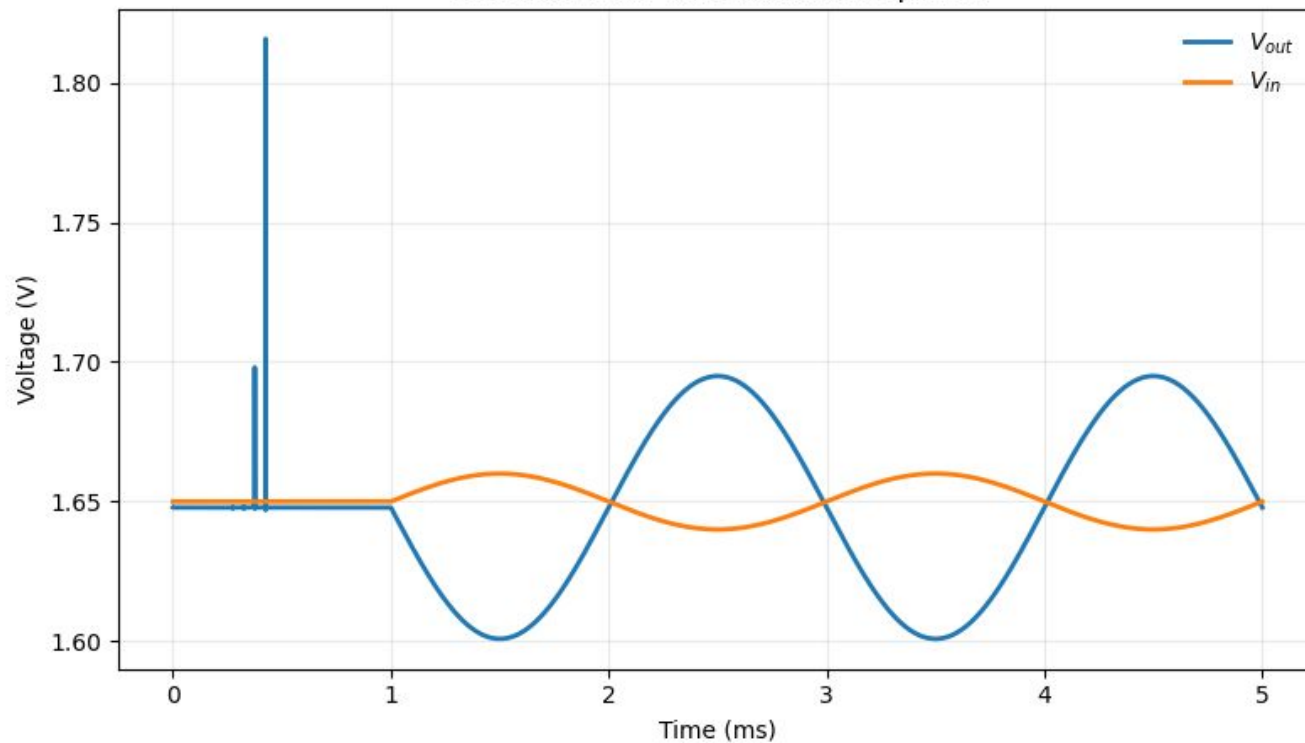
Gain

4.942105

V/V

13.88 dB

Five Transistor 4X Transient Response



Gain

4.704538

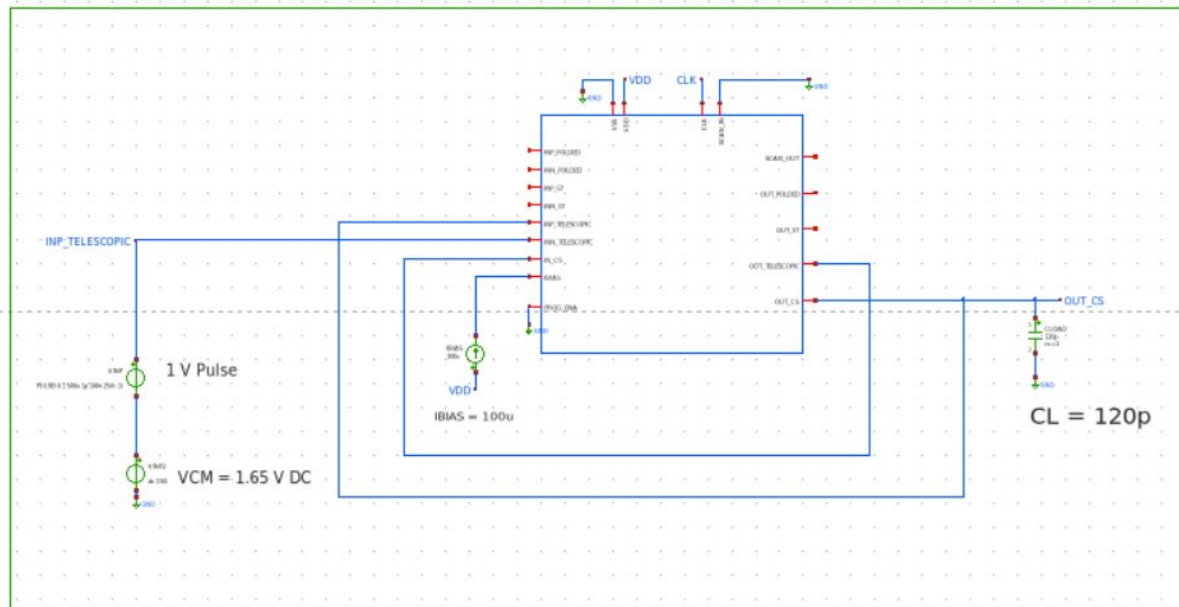
V/V

13.45 dB

Four Topology MOSbius as a Two-Stage Amplifier

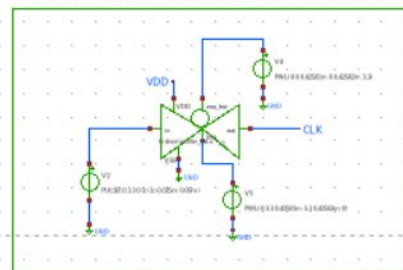
IMPLEMENTATION

Configure the telescopic cascode in 1x mode and the common source stage in 1x mode as a unity gain amplifier, stepping the input the input by 1 V while driving a 120pF capacitive load without a compensation network.



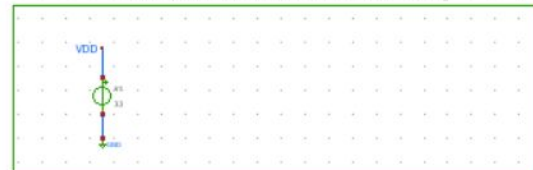
CLOCK GENERATION

Artificially generate a clock signal and ensure it turns off after around 425ms, which is when the scan in signal has fully propagated through the scan chain

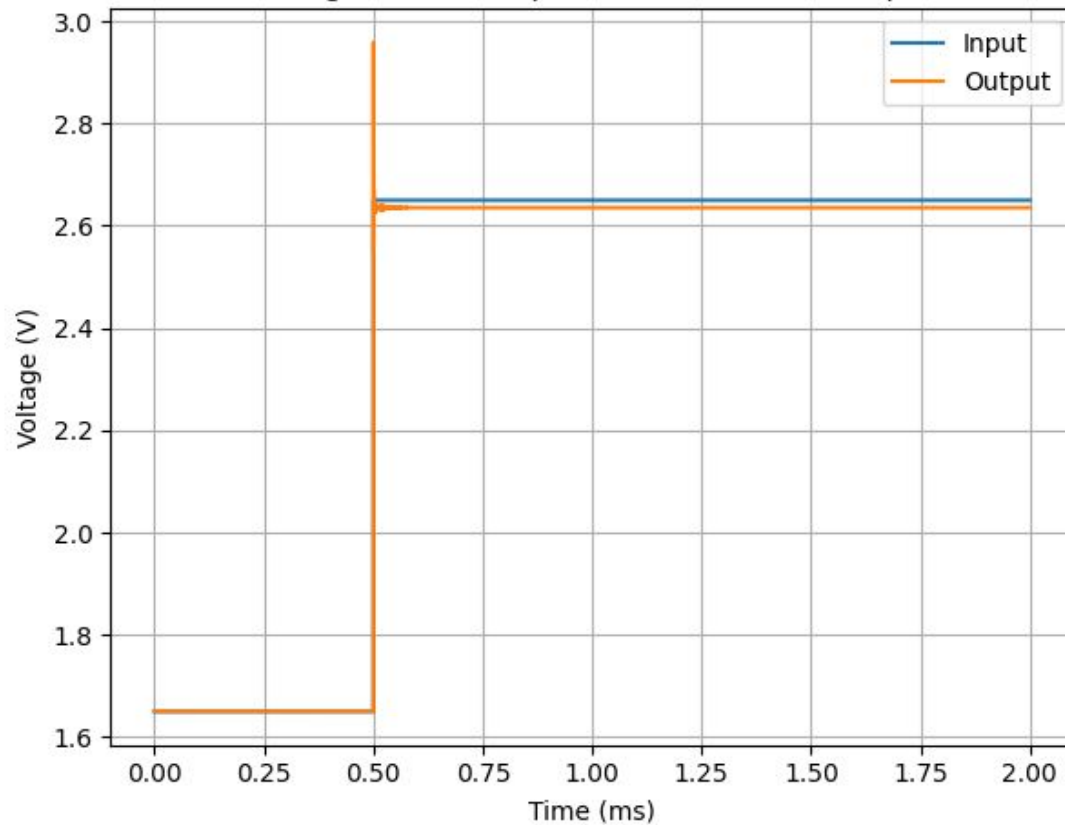


VDD, GND, SCAN INPUT

Scan chain input artificially generated to enable telescopic cascode 1x and common source 1x sizing

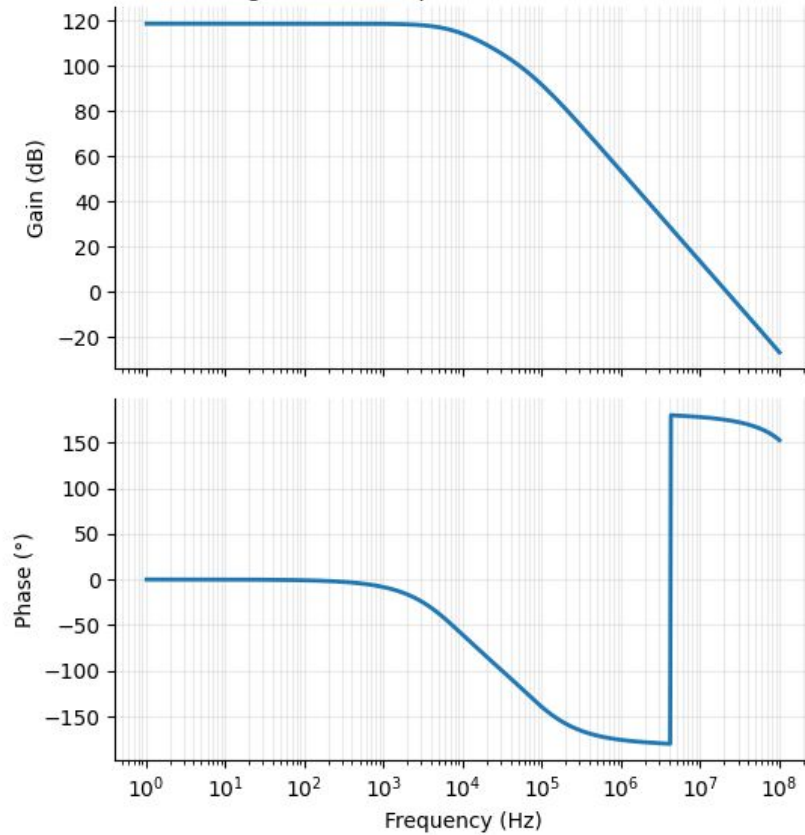


Two Stage 1X Telescopic 1X CS Transient Response



Phase Margin = 37.16°

Two-Stage 1X Telescopic Cascode 1X CS Bode Plot



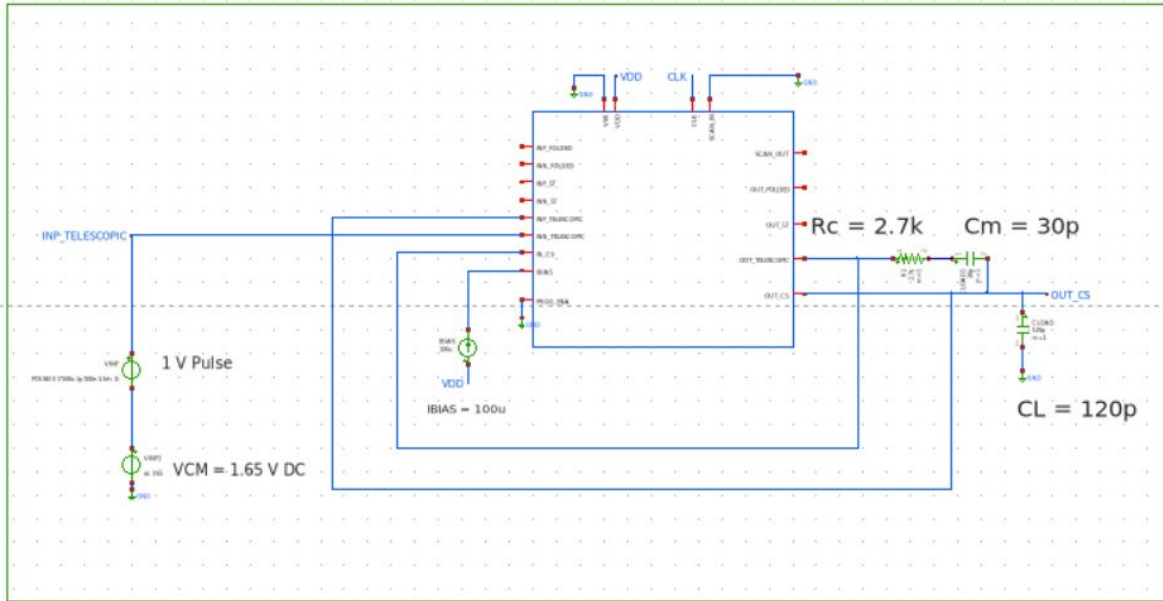
Gain = (118.62 dB)
PM = 354.6°
GBW = 21.88 MHz

Two-Stage Compensation

Users can also implement a Miller capacitor and compensation resistor to stabilize their amplifier configuration.

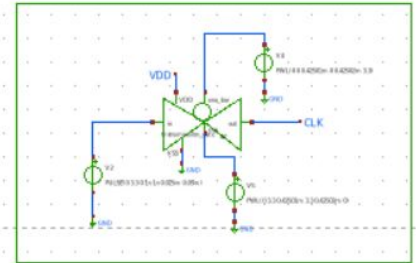
IMPLEMENTATION

Configure the telescopic cascode in 1x mode and the common source stage in 1x mode as a unity gain amplifier, stepping the input the input by 1 V while driving a 120pF capacitive load without a compensation network.



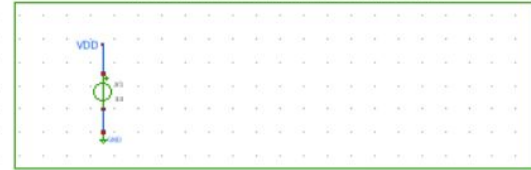
CLOCK GENERATION

Artificially generate a clock signal and ensure it turns off after around .425ms, which is when the scan in signal has fully propagated through the scan chain

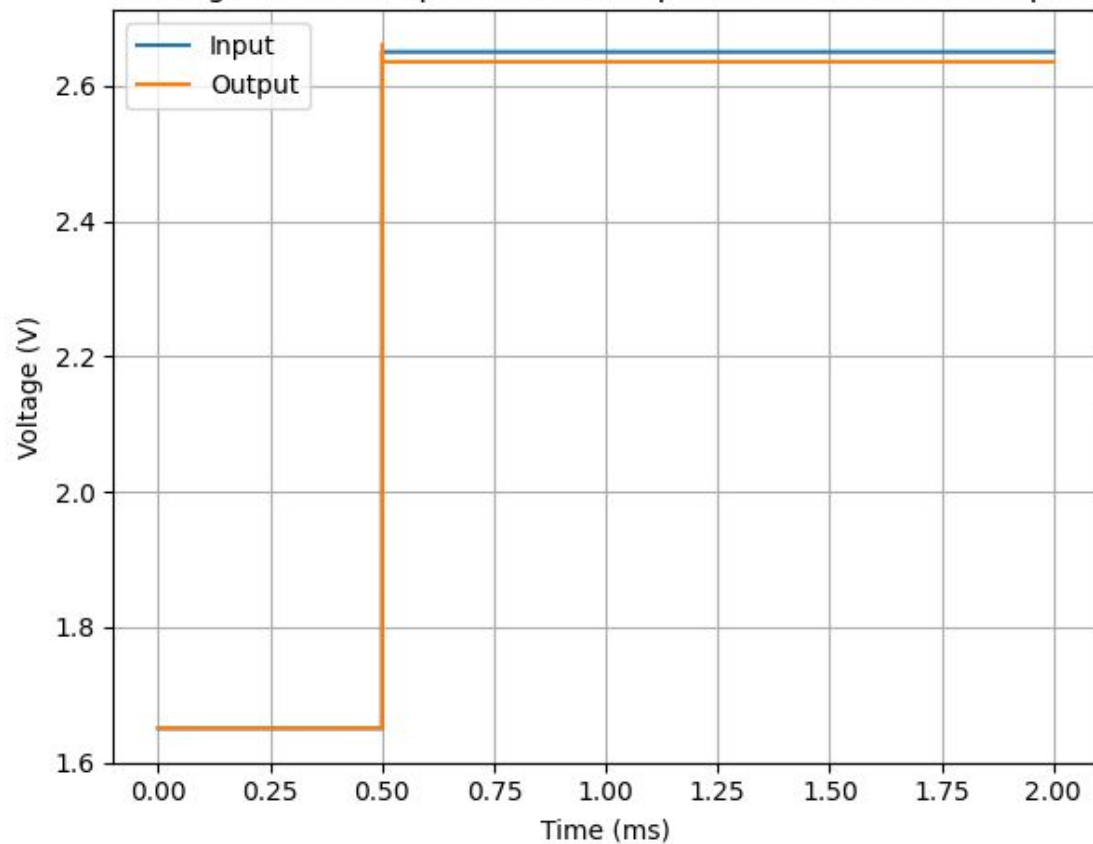


VDD, GND, SCAN INPUT

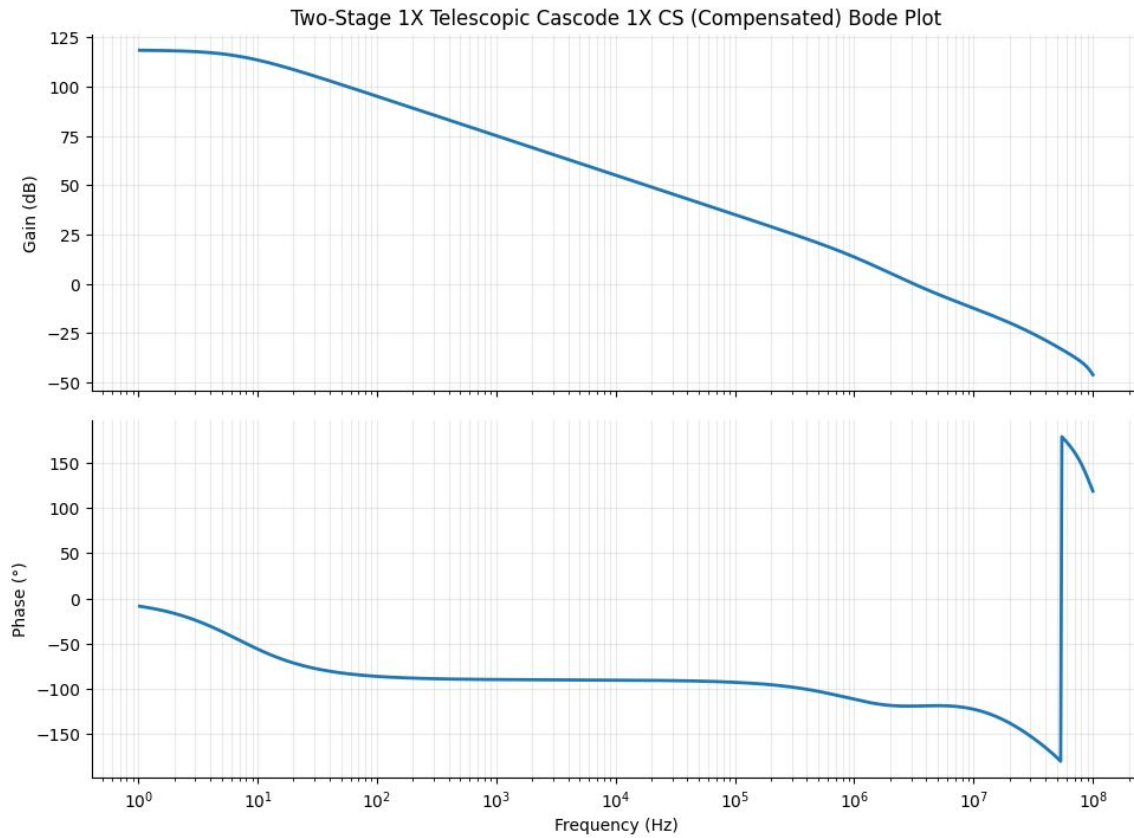
Scan chain input artificially generated to enable telescopic cascode 1x and common source 1x sizing



Two Stage 1X Telescopic 1X CS (Compensated) Transient Response



Phase Margin = 68.13°



Full Verification Results:

Folded Cascode:

<https://docs.google.com/spreadsheets/d/1ste8NNTPVhHXfTPsn5K2P177Z7tUey5MOiy1HPbO30U/edit?usp=sharing>

Telescopic Cascode:

https://docs.google.com/spreadsheets/d/1UWE3XYKk2p_d3-MaK_v5p6VSCiXoplRrxV4e-1WU_l0/edit?usp=sharing

5-Transistor OTA:

https://docs.google.com/spreadsheets/d/12NohRTnEUuWv6hk_VIJNx9IA-sGq2unuuVgBkl8fErU/edit?usp=sharing